CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Quality Assurance)

University Theory Examination March-April 2018

PHQA001 MODERN ANALYTICAL TECHNIQUES

Date: 26.03.2018, Monday Time: 10.00 a.m. to 01.00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1	**A)**	What is HPLC? Write a note on detectors used in HPLC.	**05**
	B)	Describe in brief advantages, disadvantages and application of HPTLC.	**05**
Q.2	**A)**	Give brief overview about different types of LC based on mechanism of separation.	**05**
	B)	Write a note on Gas chromatography.	**05**
		OR	
Q.2	**A)**	What is MS? Enlist various ionizer, analyzers and detectors used in MS.	**05**
	B)	Give brief account on fragmentation rules.	**05**
Q.3		**Answer the following (Any three)**	**15**
	(A)	Differentiate between HPLC and HPTLC.	
	(B)	Write a note on DTA.	
	(C)	Write a note on electrophoresis.	
	(D)	Enlist various detectors used in GC and write in brief about any one.	

SECTION-II

Q.4	**(A)**	Discuss the theory of UV Visible spectroscopy.	**05**
	(B)	Write note on Monochromators used in UV-Visible spectrophotometer.	**05**
Q.5	**(A)**	Disucss the applications of UV spectroscopy with respect to Multicomponent analysis.	**05**
	(B)	Discuss instrumentation of Spectrofluorimeter.	**05**
		OR	
Q.5		Explain Chemical shift and spin spin coupling in NMR.	**10**
Q.6		**Answer the following (Any three)**	**15**
	(A)	Differntiate Proton NMR and 13C NMR.	
	(B)	Sampling techniques in FT-IR.	
	(C)	Applications of X-Ray diffraction.	
	(D)	Brief note on ELISA.	
